Topic 1: Significant Figures **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) A student records four measurements in a lab notebook. State the number of significant figures in each.

(a) 0.00420 kg (b) 1200 m (c) 1200. m (d) 3.007×10^4 s

1a

1b

1c

1d

2) A digital balance displays the mass of a copper sample as 25.60 g. (a) How many significant figures does this reading have? (b) Rewrite the mass in scientific notation, keeping the same number of significant figures.

2a

2b

3) Round each value to three significant figures.

(a) 45,678 m (b) 0.0012349 kg (c) 8.9995 s

3a

3b

3c

4) A student adds three water samples to one beaker: 25.2 mL, 3.45 mL, and 100.1 mL. What is the total volume, expressed with the correct number of significant figures?

4

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5) An empty beaker has a mass of 47.135 g. After sand is poured in, the beaker and sand together have a mass of 156.2 g. What is the mass of the sand, expressed with the correct number of significant figures?

5

6) A rectangular lab bench measures 1.83 m long and 0.762 m wide. Calculate the area of the bench top with the correct number of significant figures.

6

7) A car travels 145.6 km in 2.5 h. What is its average speed in km/h, expressed with the correct number of significant figures?

7

8) A metal cylinder has a height of 12.5 cm and a diameter of 3.20 cm. Find its volume in cm^3. Use $V = \pi r^2 h$, and remember that π and the 2 used to find the radius are exact and do not limit the significant figures.

8

9) A hiker walks 3.45 km along a trail, stops to rest, and then walks another 1.2 km. The entire trip takes 1.50 h. Find the total distance and the hiker's average speed in km/h. Apply the addition rule first, then the division rule, and report both answers with the correct significant figures.

9

10) Two students measure the length of the same table. One reports 2 m and the other reports 2.000 m. Explain what the difference between these two results tells you about the measuring tools each student used, and why a physicist would not treat the two results as equivalent.

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